

STUART POWELL FIELD, KY, USA

WMO: 720448

Lat: 37.578N

Lon: 84.770W

Elev: 1024

StdP: 14.16

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 00144

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.8	13.6	0.1	5.7	13.8	3.5	6.8	15.9	23.3	42.4	20.6	45.0	4.4	330	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	91.2	72.6	90.0	72.6	87.9	72.5	76.6	86.7	75.5	84.9	74.4	83.3	6.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.1	127.8	81.1	72.5	124.9	80.1	72.1	123.2	79.6	40.6	87.0	39.5	85.7	38.6	82.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.0	16.8	14.5	-0.4	94.2	8.2	4.1	-6.3	97.2	-11.1	99.6	-15.7	101.9	-21.7	104.9		
			0.1	78.5	8.6	1.1	-6.1	79.3	-11.1	79.9	-15.9	80.5	-22.2	81.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.8	32.5	37.5	45.6	56.1	65.5	73.5	74.8	74.3	68.8	56.4	44.3	39.1
	DBStd	17.27	11.54	11.77	10.72	9.26	8.03	5.54	4.60	4.76	7.01	9.02	9.28	9.42
	HDD50	1798	550	374	212	45	2	0	0	0	0	37	218	360
	HDD65	4539	1007	770	602	289	98	8	2	2	43	294	622	802
	CDD50	3914	7	25	77	227	483	705	768	754	563	235	47	23
	CDD65	1181	0	0	2	21	114	263	305	291	156	27	2	0
	CDH74	10356	0	1	23	241	970	2382	2539	2511	1419	264	6	0
	CDH80	3715	0	0	0	33	259	915	942	968	533	65	0	0
Wind	WSAvg	5.3	7.1	7.0	6.7	6.2	4.9	4.7	3.4	3.2	3.6	4.8	5.4	6.2
Precipitation	PrecAvg	48.3	4.2	3.3	4.6	3.7	5.1	4.8	4.5	3.5	3.2	3.0	3.0	4.3
	PrecMax	56.7	6.4	6.8	13.2	7.2	11.1	10.5	9.4	5.6	7.6	5.6	5.5	10.9
	PrecMin	37.2	1.9	1.1	1.6	1.4	1.5	1.7	0.8	1.1	0.7	0.7	1.0	2.1
	PrecStd	6.3	1.1	1.4	2.6	1.4	2.4	2.2	1.9	1.1	2.0	1.7	1.1	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.3	70.2	78.5	83.8	90.4	96.7	92.9	95.1	94.6	86.3	73.0	66.2
		MCWB	56.7	60.6	61.8	60.6	70.6	71.1	73.8	73.2	71.5	70.9	57.7	59.1
	2%	DB	59.1	64.3	72.3	80.8	85.8	90.7	90.3	90.7	89.9	81.0	68.1	63.0
		MCWB	54.0	56.3	59.0	62.6	69.7	73.1	73.9	72.2	70.4	66.2	58.0	57.7
	5%	DB	54.8	61.2	67.6	76.9	82.1	88.4	88.2	88.4	86.2	76.7	63.1	58.9
		MCWB	50.3	54.1	56.5	60.8	68.0	72.7	74.0	72.2	70.6	63.5	55.1	54.8
	10%	DB	50.4	56.5	63.1	72.5	80.7	85.9	85.9	86.0	81.9	71.8	59.4	54.7
		MCWB	46.5	50.5	53.2	59.3	67.2	71.8	73.5	72.1	68.0	61.8	52.1	51.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.0	61.9	63.8	68.0	73.2	77.1	78.1	77.5	75.2	72.5	63.6	61.5
		MCDB	62.6	67.7	75.2	77.9	84.7	87.7	86.8	89.1	85.0	82.1	69.0	63.6
	2%	WB	55.3	59.4	60.9	65.8	71.5	75.6	77.0	76.2	73.7	69.8	60.7	58.6
		MCDB	58.2	63.9	69.0	76.1	81.5	86.3	86.8	86.4	83.9	75.5	64.7	62.4
	5%	WB	51.6	55.5	58.5	63.3	70.2	74.3	75.9	74.9	72.3	66.7	56.6	55.0
		MCDB	54.0	60.3	65.0	71.5	80.3	84.6	84.8	84.0	81.8	72.5	60.9	57.8
	10%	WB	46.4	52.3	55.3	61.2	68.6	73.0	74.5	73.6	70.8	63.4	53.3	52.3
		MCDB	48.6	55.7	61.2	69.0	78.0	82.6	82.6	81.7	78.7	69.4	58.7	55.0
Mean Daily Temperature Range	5% DB	MDBR	16.3	17.8	19.8	23.4	20.8	21.2	19.6	21.3	21.8	21.3	19.2	15.3
		MCDBR	20.9	22.9	26.1	28.2	24.2	24.4	22.1	24.9	26.3	25.7	23.0	18.9
		MCWBR	16.7	16.3	15.4	14.2	10.8	9.6	8.9	9.3	9.9	12.2	14.0	15.6
	5% WB	MCDBR	18.9	21.2	21.3	22.0	21.3	20.8	19.1	20.0	21.5	21.2	18.0	17.9
		MCWBR	17.0	17.1	14.1	13.2	10.0	9.4	8.7	8.6	9.4	11.7	13.8	15.7
Clear-Sky Solar Irradiance	taub	0.307	0.320	0.350	0.392	0.442	0.452	0.480	0.478	0.411	0.356	0.321	0.301	
	taud	2.477	2.428	2.394	2.275	2.174	2.184	2.120	2.116	2.276	2.427	2.487	2.505	
	Ebn at Noon	279	289	289	282	268	265	256	253	265	272	272	274	
	Edh at Noon	26	31	35	42	47	47	49	48	39	30	25	23	
All-Sky Solar Radiation	RadAvg	624	831	1186	1578	1765	1968	1846	1739	1443	1094	765	545	
	RadStd	71	121	105	118	164	104	97	94	152	132	81	53	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+220	+137

Nomenclature: See separate page